Synoptic Question 31: The Physics of a Washing Machine Spin Cycle

Reading Passage

Read the following article and then answer the questions that follow.

An automatic washing machine uses mechanical agitation and thermal energy to clean clothes. A key part of the process is the spin cycle, designed to remove excess water through rapid rotation. The inner drum, containing the wet clothes, is accelerated to a high angular velocity. Water is forced out through perforations in the drum by the large centripetal force required to keep the water moving in a circle.

The entire drum assembly (including the motor and outer tub) is typically suspended within the machine's casing by a system of springs and dampers (shock absorbers), as shown schematically in Fig. 31.1. This suspension allows the assembly to vibrate during the spin cycle, particularly if the load of clothes is unevenly distributed (unbalanced).

Every mechanical system has one or more natural frequencies at which it tends to vibrate with large amplitude if driven at that frequency. This phenomenon is called **resonance**. If the rotational frequency of the drum during the spin cycle matches a natural frequency of the suspended assembly, violent vibrations can occur, potentially damaging the machine or causing excessive noise. Washing machine designs incorporate suspension systems with specific stiffness and damping characteristics to mitigate these vibrations and avoid resonance at the main operating speeds. Often, the machine accelerates rapidly through lower speeds that might correspond to natural frequencies. Damping, provided by the shock absorbers, dissipates vibrational energy as heat.

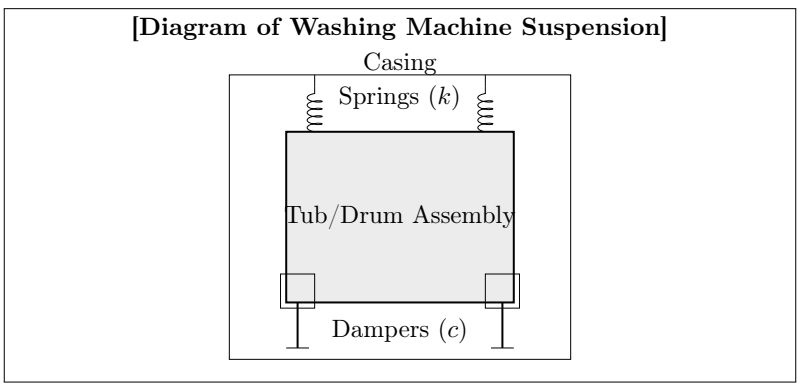

Figure 1: Simplified schematic of drum assembly suspension.

Data:

Data for a typical front-loading washing machine during its spin cycle:

Table 1: Data for Washing Machine Spin Cycle Analysis

Parameter	**Value**
Inner radius of drum, r	0.25 m
Total mass of suspended assembly (tub, drum, motor, wet clothes), m_{tub}	65.0 kg
Total effective spring constant of suspension, k_{susp}	2.80×10^4 N m^{-1}
Maximum spin speed, $\omega_{max,rpm}$	1400 rpm
Time taken to accelerate from rest to max spin speed, t_{accel}	8.0 s
Moment of inertia of loaded drum (approximated), I_{drum}	1.80 kg m^2
Resistive torque during spin (friction, damping), τ_{resist}	2.50 Nm
Efficiency of motor, η_{motor}	70%
Mass of unbalanced load clump, m_{unb}	0.50 kg

Questions

(a) Define **natural frequency** and **resonance**.

(b) The suspended tub assembly can oscillate vertically on the springs.

  (i) Calculate the natural frequency f_n (in Hz) of vertical oscillation for the loaded tub assembly.

  (ii) Convert the maximum spin speed (1400 rpm) into a rotational frequency f_{spin} in Hz.

  (iii) Compare f_n and f_{spin}. Explain why it is crucial for a washing machine design that these two frequencies are significantly different.

  (iv) The machine accelerates through various speeds before reaching f_{spin}. Suggest why the machine might be programmed to accelerate particularly quickly through the speed range around f_n.

(c) Consider the drum rotating at its maximum spin speed.

  (i) Calculate the angular velocity ω_{max} in rad s^{-1}.

  (ii) Calculate the magnitude of the centripetal acceleration experienced by clothes at the drum's inner radius. Express your answer in terms of g.

(d) During acceleration to maximum spin speed:

  (i) Assuming constant angular acceleration α, calculate its value.

  (ii) Calculate the net torque τ_{net} required to produce this acceleration.

  (iii) Calculate the total torque τ_{motor} that the motor must provide during acceleration, considering the resistive torque.

(e) Consider the energy required during the spin cycle.

  (i) Calculate the rotational kinetic energy stored in the drum at maximum spin speed.

  (ii) Calculate the mechanical power P_{mech} the motor must provide to maintain the maximum spin speed against the resistive torque.

  (iii) Calculate the electrical power P_{elec} consumed by the motor to maintain this speed.

(f) An unbalanced load occurs when clothes gather unevenly, modelled here as a 0.50 kg mass clump stuck to the inside radius of the drum.

  (i) Calculate the magnitude of the unbalanced force generated by this mass rotating at the maximum spin speed.

  (ii) Describe qualitatively how the suspension system (springs and dampers) responds to this rotating unbalanced force. Explain the role of the dampers.

Solutions

(a) **Natural Frequency (f_n):** The frequency at which a system oscillates when disturbed from its equilibrium position and allowed to oscillate freely (without any driving or damping forces).
Resonance: The phenomenon that occurs when a system is driven by a periodic external force at a frequency equal or very close to its own natural frequency, resulting in a large increase in the amplitude of oscillation.

(b) (i) For a mass-spring system, $f_n = \frac{1}{2\pi}\sqrt{\frac{k_{susp}}{m_{tub}}}$.
$f_n = \frac{1}{2\pi}\sqrt{\frac{2.80\times10^4}{65.0}} = \frac{1}{2\pi}\sqrt{430.8} = 3.30$ Hz.

(ii) $f_{spin} = 1400 \text{ rpm} \times \frac{1 \text{ min}}{60 \text{ s}} = 23.33 \text{ Hz} \approx 23.3$ Hz.

(iii) $f_n(3.30 \text{ Hz}) \ll f_{spin}(23.3 \text{ Hz})$. It is crucial that the main operating frequency (f_{spin}) is far from the natural frequency (f_n) to avoid resonance, which would cause large amplitude, noisy, and potentially damaging vibrations.

(iv) Accelerating quickly through the speed range near f_n (approx 3.3 Hz $\times$ 60 $\approx$ 200 rpm) minimizes the time spent driving the system at its resonant frequency. This prevents the amplitude of vibrations from building up significantly during start-up or slow-down.

(c) (i) $\omega_{max} = f_{spin} \times 2\pi = 23.33 \times 2\pi = 146.6 \text{ rad s}^{-1} \approx 147 \text{ rad s}^{-1}$.

(ii) $a_c = r\omega_{max}^2 = (0.25)(146.6)^2 = 5373 \text{ m s}^{-2}$.
In terms of g: $\frac{a_c}{g} = \frac{5373}{9.81} = 548$. So, $a_c \approx 550g$.

(d) (i) $\alpha = \frac{\omega_f - \omega_i}{t_{accel}} = \frac{146.6-0}{8.0} = 18.33 \text{ rad s}^{-2} \approx 18.3 \text{ rad s}^{-2}$.

(ii) $\tau_{net} = I_{drum}\alpha = (1.80)(18.33) = 33.0$ Nm.

(iii) The motor must provide torque to accelerate the drum (τ_{net}) AND overcome friction (τ_{resist}).
$\tau_{motor} = \tau_{net} + \tau_{resist} = 33.0 + 2.50 = 35.5$ Nm.

(e) (i) $KE_{rot} = \frac{1}{2}I_{drum}\omega_{max}^2 = \frac{1}{2}(1.80)(146.6)^2 = 19343 \text{ J} \approx 19.3 \text{ kJ}$.

(ii) To maintain constant speed, the motor torque must equal the resistive torque.
$P_{mech} = \tau_{resist}\omega_{max} = (2.50)(146.6) = 366.5 \text{ W} \approx 367 \text{ W}$.

(iii) $P_{elec} = \frac{P_{mech}}{\eta_{motor}} = \frac{366.5}{0.70} = 523.6 \text{ W} \approx 524 \text{ W}$.

(f) (i) The unbalanced mass m_{unb} moves in a circle of radius r. It requires a centripetal force $F_{unb} = m_{unb}r\omega_{max}^2$.
$F_{unb} = (0.50)(0.25)(146.6)^2 = 2687 \text{ N} \approx 2690 \text{ N}$.
This rotating force is transmitted to the tub assembly.

(ii) The rotating unbalanced force acts as a periodic driving force on the suspended tub assembly, causing it to vibrate. The springs provide the restoring force for this vibration. The dampers (shock absorbers) exert a force that opposes the velocity of the vibration, dissipating the vibrational energy as heat. This limits the amplitude of the vibrations, especially if the driving frequency were close to the natural frequency, preventing excessive shaking of the machine.

Cambridge International AS & A Level Physics

Paper 4 Style – Variant 2 (Higher Difficulty)

Structured and Open-Ended Questions

Instructions to candidates

- Answer all questions.
- Write your working clearly in the spaces provided.
- Use an appropriate number of significant figures.
- Where assumptions are made, state them explicitly.

Data: Gravitational field strength $g = 9.81\,\text{m s}^{-2}$.

Total for Variant 2: 40 marks

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1 A washing machine drum of radius 0.24 m accelerates from rest to 1200 rev min^{-1} in 7.0 s. The wet laundry is modeled as a ring of mass 3.8 kg concentrated near the drum wall.

1.1 Show that the final angular speed is approximately 1.26×10^2 rad s^{-1}. [2]

1.2 Calculate the angular acceleration of the laundry. [2]

1.3 Determine the increase in rotational kinetic energy of the laundry. [3]

1.4 The motor-drivetrain efficiency during this interval is 62%. Estimate the electrical energy drawn from the mains. [2]

1.5 State one reason why your result in 1.4 may differ from an experimental value. [1]

2 During spin, load imbalance is modeled as an effective point mass 0.35 kg at radius 0.20 m rotating with angular speed ω. The tub and drum assembly has mass 28 kg and is mounted on springs and dampers.

2.1 Write an expression for the amplitude of the periodic forcing caused by the imbalance. [2]

2.2 Explain why operating close to resonance can cause severe vibration. [2]

2.3 Suggest two engineering methods to reduce transmitted vibration and, for each method, state one trade-off. [4]

2.4 A designer increases damping substantially. Discuss the effect on:
(i) passing through resonance during spin-up, and
(ii) steady operation at high speed. [2]

3 In one programme, 18 L of water is heated from 18°C to 55°C. Assume density of water is 1.0 kg L^{-1} and specific heat capacity is 4180 J kg^{-1}K^{-1}.

3.1 Calculate the thermal energy gained by the water. [3]

3.2 The heater power is 2.0 kW and 85% of electrical energy reaches the water. Estimate the minimum heating time. [3]

3.3 State two reasons why the actual heating time may be longer. [2]

3.4 Suggest one method to reduce energy use while maintaining cleaning performance, and explain the physics. [2]

4 During extraction, water is expelled from fabric as outward inertial effects increase with spin speed.

4.1 For a drum radius of 0.22 m and speed of 1000 rev min^{-1}, calculate the centripetal acceleration in:
(i) m s^{-2} and
(ii) units of g. [4]

4.2 Explain how an increase in spin speed affects residual moisture content. [2]

4.3 Suggest one limitation of the rule "higher spin speed always gives better design performance". [2]

4.4 Explain how drum perforation design and fabric type influence extraction effectiveness. [2]

Sample Solutions (Variant 2, CIE P4 Style)

1.1 $1200\,\text{rev min}^{-1} = \dfrac{1200}{60} = 20\,\text{rev s}^{-1}$,

$$\omega = 2\pi f = 2\pi(20) = 40\pi \approx 1.26 \times 10^2\,\text{rad s}^{-1}.$$

1.2

$$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{125.7 - 0}{7.0} \approx 18.0\,\text{rad s}^{-2}.$$

1.3

$$I = mr^2 = 3.8(0.24)^2 = 0.2189\,\text{kg m}^2,$$

$$\Delta E_k = \frac{1}{2}I\omega^2 = \frac{1}{2}(0.2189)(125.7)^2 \approx 1.73 \times 10^3\,\text{J}.$$

1.4

$$E_{\text{electrical}} = \frac{\Delta E_k}{0.62} \approx 2.79 \times 10^3\,\text{J}.$$

1.5 Example: additional losses due to bearing friction, air resistance, or non-rigid redistribution of clothes.

2.1 $F_0 = m_u r\omega^2$.

2.2 At resonance, response amplitude is greatly magnified so displacement and transmitted forces rise sharply, increasing noise and structural stress.

2.3 Example answers:

- Increase active balancing control; trade-off: longer cycle or added control complexity.
- Optimise suspension stiffness/damping; trade-off: better stability in one regime may worsen force transmission in another.

2.4 More damping reduces peak resonance amplitude during run-up, but can increase dissipative losses and may transmit more force away from resonance.

3.1

$$m = 18\,\text{kg}, \quad \Delta T = 37\,\text{K}, \quad Q = mc\Delta T = 18(4180)(37) \approx 2.78 \times 10^6\,\text{J}.$$

3.2

$$P_{\text{to water}} = 0.85(2.0 \times 10^3) = 1.70 \times 10^3\,\text{W},$$

$$t = \frac{Q}{P} \approx \frac{2.78 \times 10^6}{1.70 \times 10^3} \approx 1.63 \times 10^3\,\text{s} \approx 27.2\,\text{min}.$$

3.3 Example reasons: heating of drum/tub material; heat loss to surroundings; thermostat cycling.

3.4 Example: use lower wash temperature with longer agitation and suitable detergent chemistry to reduce thermal energy demand while maintaining cleaning action.

4.1

$$\omega = 2\pi\left(\frac{1000}{60}\right) = 104.7\,\text{rad s}^{-1},$$

$$a_c = \omega^2 r = (104.7)^2(0.22) \approx 2.41 \times 10^3\,\text{m s}^{-2},$$

$$\frac{a_c}{g} = \frac{2.41 \times 10^3}{9.81} \approx 2.46 \times 10^2 \approx 246\,g.$$

4.2 Higher spin speed increases outward inertial effects, generally lowering residual moisture.
4.3 Excessive speed can increase wear, vibration, noise, and mechanical stress with diminishing extraction gains.
4.4 Perforation geometry affects drainage path and flow resistance; fabric weave/absorbency determines how readily water migrates and is expelled.

Mark Scheme (Variant 2, CIE P4 Style)

Total: 40 marks

Part	Marking points	Marks
1.1	Converts rpm to rad s^{-1} correctly and gives $\omega \approx 126\,\text{rad s}^{-1}$.	2
1.2	Uses $\alpha = \Delta\omega/\Delta t$ correctly with valid value and unit.	2
1.3	Uses $I = mr^2$ and $\Delta E_k = \frac{1}{2}I\omega^2$ with correct method and unit.	3
1.4	Applies efficiency relation correctly.	2
1.5	One valid physical reason for discrepancy.	1
2.1	States $F_0 = m_u r\omega^2$.	2
2.2	Correct qualitative resonance explanation.	2
2.3	Two valid engineering methods + trade-off for each.	4
2.4	Balanced discussion of increased damping effects.	2
3.1	Correct $Q = mc\Delta T$ substitution and result.	3
3.2	Correct effective power and time estimate.	3
3.3	Two valid reasons for longer real heating time.	2
3.4	One justified energy-saving strategy using physics.	2
4.1	Correct ω, a_c, and conversion to g.	4
4.2	Correct trend of moisture content with speed and reason.	2
4.3	One valid design limitation at high rpm.	2
4.4	Explains roles of drum perforation and fabric properties.	2

Examiner notes:

- Award method marks where physics setup is correct even with arithmetic slips.
- Accept equivalent models and valid alternative reasoning.
- Penalise missing units once per part unless ambiguity affects interpretation.